

# Shaurya Patel

6335 Thunderbird Crescent, Vancouver, BC V6T 2G9 • +1(236) 516-7246

Github: shauryapatel1995 • E-mail: spatel27@student.ubc.ca

## Research Interests

---

Operating Systems, Systems Performance.

## Education

---

**University of British Columbia**, *Ph.D. in ECE* Vancouver, Canada (Sep 2021 – May 2027 (expected))

*Advisors: Dr. Alexandra Fedorova and Dr. Margo Seltzer*

**University of Massachusetts**, *Masters of Science in Computer Science* Amherst, MA (Sep 2019 – May 2021)

**Nirma Institute of Technology**, *Bachelor of Science in Computer Science and Engineering* Ahmedabad, India (Aug 2013 – May 2017)

## Peer-reviewed Publications

---

- (1) Agrawal, S., **Patel, S.**, Pham, L., Stevinson, A., Karimalis, I., Lefevre, H., Mehta, A., Achermann, R., Seltzer, M., “Exploring the Design Space of Isolation Mechanisms,” Proceedings of Workshop on Programming Languages and Operating Systems (PLOS), 2026.
- (2) Tzafrir, D., **Patel, S.**, Nider, J., Fedorova, A., “PHASE: A Platform for Hardware and Software Evaluation of Memory Tiering,” Proceedings of Workshop on Disruptive Memory Systems (DIMES), 2026.
- (3) **Patel, S.**, Yang, S., Wang, Y., Wu, K., Fedorova, A., Seltzer, M., Keeton, K., “MDK: rethinking the data center memory reclamation problem,” Proceedings of Operating systems design and implementation (OSDI), 2026.
- (4) Agrawal, S., **Patel, S.**, Pham, L., Stevinson, A., Karimalis, I., Lefevre, H., Mehta, A., Achermann, R., Seltzer, M., “Comparing Isolation Mechanisms with OSmosis,” Proceedings of Workshop on Programming Languages and Operating Systems (PLOS), 2025.
- (5) Jalalian, S., **Patel, S.**, Rezaei, M., Seltzer, M., Fedorova, A., “ExtMem: Enabling Application-Aware Virtual Memory Management for Data-Intensive Applications,” Proceedings of USENIX Annual Technical Conference (ATC), 2024.
- (6) Cao, X., **Patel, S.**, Lim, S., Han, X., Pasquier, T., “FetchBPF: Customizable Prefetching Policies in Linux with eBPF,” USENIX Annual Technical Conference (ATC), 2024.
- (7) **Patel, S.**, Agrawal, S., Fedorova, A., Seltzer, M., “CHERI-picking: Leveraging Capability Hardware for Prefetching,” Proceedings of Workshop on Programming Languages and Operating Systems (PLOS), 2023.
- (8) **Patel, S.**, Liu, T., Guan, H., “FreeLunch: Compression-based GPU Memory Management for Convolutional Neural Networks,” Proceedings of IEEE/ACM Workshop on Memory Centric High Performance Computing (MCHPC), 2021.

## Professional Experience

---

**Google**, *Student Researcher*

Sunnyvale, USA and Remote (May 2024 – April 2026)

|                                                  |                                          |
|--------------------------------------------------|------------------------------------------|
| <b>Google</b> , <i>Software Developer Intern</i> | Vancouver, Canada (Sep 2022 – Dec 2022)  |
| <b>Google</b> , <i>Software Developer Intern</i> | Mountain View, USA (May 2020 – Aug 2020) |
| <b>Morgan Stanley</b> , <i>Senior Associate</i>  | Mumbai, IN (Aug 2017 – Jul 2019)         |
| <b>Morgan Stanley</b> , <i>Intern</i>            | Mumbai, IN (Jan 2017 – Jul 2017)         |

## Awards and Grants

---

|                                                            |      |
|------------------------------------------------------------|------|
| • OSDI Student Grant                                       | 2026 |
| • HotOS Student Grant                                      | 2025 |
| • SOSP Student Grant                                       | 2023 |
| • Eurosys submission chair Grant                           | 2023 |
| • OSDI and PLMW Student Grant                              | 2020 |
| • Morgan Stanley Global Tech Excellence award              | 2018 |
| • Morgan Stanley Outstanding Delivery award                | 2018 |
| • Morgan Stanley Best Project award                        | 2018 |
| • Regional Mathematical Olympiad scholarship (3rd in city) | 2009 |
| • Outstanding Student of the Year award                    | 2007 |

## Service

---

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| • PC meeting volunteer, SOSP                                        | 2024      |
| • Submission chair, EUROSYS                                         | 2023      |
| • Vice president Social External, ECEGSA                            | 2022–2023 |
| • Student volunteer, SPLASH conference, Online                      | 2020      |
| • President, Computer society of India, Nirma University, Ahmedabad | 2016      |
| • Core planning team, NUTech symposium, Ahmedabad                   | 2016      |
| • Mathematics teacher, Grade 5, Numeracy Hope, Ahmedabad            | 2015      |

## Undergraduate Research Mentoring

---

|                                                                                                                               |              |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|
| • <b>Paul Van Raamsdonk</b> (UBC) — “SLObro: tuning data center applications the right way”                                   | 2026–current |
| • <b>Sissi Qi</b> (UBC) — “SLObro: tuning data center applications the right way”                                             | 2025–current |
| • <b>Kevin Ding</b> (UBC) — “Coffee: Going beyond the next hit for OS prefetching”                                            | 2024–current |
| • <b>Rhys Byers</b> (UBC) — “Coffee: Going beyond the next hit for OS prefetching”                                            | 2024–current |
| • <b>Mabel Wang</b> (UBC) — “Coffee: Going beyond the next hit for OS prefetching”<br><i>Now: MASc at UBC ECE</i>             | 2025–2026    |
| • <b>Jayden Lin</b> (CMU) — “Coffee: Going beyond the next hit for OS prefetching”<br><i>Now: B.Sc. in CS at CMU</i>          | 2025–2026    |
| • <b>Tom Wang</b> (UBC) — “Coffee: Going beyond the next hit for OS prefetching”<br><i>Now: Intern at Alterra</i>             | 2022–2024    |
| • <b>Yisheng Liu</b> (UBC) — “Coffee: Going beyond the next hit for OS prefetching”<br><i>Now: Ph.D. student at UWaterloo</i> | 2022–2024    |
| • <b>Victor Garvalov</b> (EPFL) — “Sampling memory access for effective page eviction”<br><i>Now: M.Sc. student at EPFL</i>   | 2023–2024    |
| • <b>Jackson Dagger</b> (UBC) — “Under pressure: how memory pressure changes patterns in swap accesses”                       | 2022–2023    |